Homework #2

Problem 1

There are 3 different processors P1, P2 and P3 with the same ISA, its clock frequencies and CPIs are as follow:

Processor	Clock Frequency	CPI
P1	2 GHz	1.5
P2	1.5 GHz	1.2
P3	3 GHz	2.0

1. Calculate the MIPS for each processor and find out which one has the best performance?
2. If we want to increase the MIPS by 30%, which will cause an increase of CPI by 15%, what's the clock frequency should be for each processor?

解:

$$\text{CPU时间} = \text{指令数} \times \text{CPI} \times \text{时钟周期长度}$$

$$\text{MIPS} = \frac{\text{指令数}}{\text{执行时间} \times 10^6} = \frac{\text{时钟频率}}{\text{CPI} \times 10^6}$$

1. $\text{MIPS}_{P1} = \frac{2\text{GHz}}{1.5 \times 10^6} = 1333.3$

$\text{MIPS}_{P2} = \frac{1.5\text{GHz}}{1.2 \times 10^6} = 1250$

$\text{MIPS}_{P3} = \frac{3\text{GHz}}{2.0 \times 10^6} = 1500$

P3 has the best performance

2. $CF'_{P1} = 1.3 \times 1.15 \times CF_{P1} = 2.99\text{GHz}$

$CF'_{P2} = 1.3 \times 1.15 \times CF_{P2} = 2.2425\text{GHz}$

$CF'_{P3} = 1.3 \times 1.15 \times CF_{P3} = 4.485\text{GHz}$

Problem 2

The table below shows the instruction type breakdown of a given application executed on 1, 2, 4, or 8 processors. Using this data, you will be exploring the speedup of applications on parallel processors.

Processors	No. Instructions per processor			CPI		
	Arithmetic	Load/Store	Branch	Arithmetic	Load/Store	Branch
1	2560	1280	256	1	4	2
2	1280	640	128	1	5	2
4	640	320	64	1	7	2
8	320	160	32	1	12	2

1.1 The table above shows the number of instructions required per processor to complete a program on a multiprocessor with 1, 2, 4, or 8 processors. What is the total number of instructions executed per processor? What is the aggregate number of instructions executed across all processors?

合计值

1.2 Given the CPI values on the right of the table above, find the total execution time for this program on 1, 2, 4, and 8 processors. Assume that each processor has a 2 GHz clock frequency.

1.3 If the CPI of the arithmetic instructions was doubled, what would the impact be on the execution time of the program on 1, 2, 4, or 8 processors? Assume that each processor has a 2 GHz clock frequency.

解. 1.1, 1.2, 1.3

Processors	Instructions per Processor	Aggregate Number of Instructions	Total Execution Time	If doubled
1	4096	4096	4.096μs	5.376μs
2	2048	4096	2.368μs	3.008μs
4	1024	4096	1.504μs	1.824μs
8	512	4096	1.152μs	1.312μs

The table below shows the number of instructions per processor core on a multicore processor as well as the average CPI for executing the program on 1, 2, 4, or 8 cores. Using this data, you will be exploring the speedup of applications on multicore processors.

Cores per Processor	Instructions per Core	Average CPI
1	1.00E+**10**	1.2
2	5.00E+**9**	1.4
4	2.50E+**9**	1.8
8	1.25E+**9**	2.6

1.4 Assuming a 3 GHz clock frequency, what is the execution time of the program using 1, 2, 4, or 8 cores?

1.5 If using a single core, find the required CPI for this core to get an execution time equal to the time obtained by using the number of cores in the table above (execution times in problem 1.4). Note that the number of instructions should be the aggregate number of instructions executed across all the cores.

解. 1.4 1.5

Processors	Execution time of the program	CPI needed
1	4s	1.2
2	2.33s	0.7
4	1.5s	0.45
8	1.08s	0.375